

Answer once, reconsider what depends on it

11 September 2026

Finding and practical outcome

A person can now use a local decision desk to answer one planning question for a stated scope, see which agent drafts can use that answer, make an exception, and identify drafts needing revalidation after a change. The prototype records the source and scope of each answer and blocks release of drafts whose registered decision versions have changed. The demonstration includes actual model interpretation failures and an explicit request-only exception.

The experiment does **not** support the more elaborate controller as a better automatic answer-reuse method. On 24 held-out MultiWOZ source dialogues, selective dependency checking and a competent semantic-memory adaptation completed the same whole projects correctly. Selective checking required 1.875 more additional questions per project, with a paired 95% bootstrap interval [1.041, 2.854]. A simple global version check achieved exactly the same outcomes as selective checking. The richer structure saved machine rereads, not demonstrated human effort. On separate authored challenges, full-history answering handled exceptions and revocation better than the structured extractor.

The useful avenue is therefore **visible, editable user decisions with ordinary change-consistency checks**, backed by a strong shared-context baseline. Do not promote the current extractor as reliable automatic authority. This is follow-up research and a working interaction prototype, not a new scheduling contribution or an amendment to the completed ICAART submission.

Research question and closest alternatives

When several agents use a person's requirements, can explicit decision scope and version checks reduce repeated clarification without spreading incorrect or stale answers? The distinguishing outcome is whether dependent work still satisfies the user's current requirements when it is released. Accurate memory retrieval alone does not establish this property, and version consistency alone does not establish that an instruction was understood correctly.

Coordinated clarification already appears in [MAC, Sections 3-5](#). [AgentAsk, Sections 3-5](#) addresses information gaps at inter-agent handoffs. [Mem0](#) and [Zep](#) already extract, update and invalidate memory. [Value of Information](#) already evaluates whether clarification is worth its communication cost. [HiLSVA](#) already combines reusable feedback, concurrent sessions and provenance. Dependency invalidation is established in truth maintenance and incremental build systems. None of those mechanisms becomes new by placing it in an agent interface.

The [literature comparison](#) identifies inspected sections, released code revisions, publication status and access limitations. Our all-current-memory selector is a Mem0-inspired **adaptation**, not a reproduction of Mem0's complete retrieval/update/graph system. It reconstructs current scalar memories from each released prefix and gives the selector all of them. Full-history answering is a separate strong alternative. No baseline receives a deliberately stale memory store or an artificially restrictive top-k limit.

Three candidates were examined. Demand admission was tested in four controlled task sets. A simple policy that lets blocked workers yield was strong, and the proposed human-aware rule made urgent outcomes worse in one case. Resumption support was retained as a change-impact view, without a separate summary-system claim. Decision reuse was selected because the source utterances, revisions and resulting local artifacts could be checked within the budget. The two adaptation checkpoints and rejected hypotheses are in [DECISIONS.md](#).

Evidence, task construction and freeze

What the source is

The application is a local travel-planning project. People can supply preferences such as location, price range and accommodation type without specialist expertise. The source is [MultiWOZ 2.4](#), pinned at 6807c1d85f547fcaae10494d26991d2d37c90a63, under its released MIT license. It contains task-elicited human Wizard-of-Oz dialogues and benchmark database records. It is not observed use of our system. The 2.4 authors corrected validation/test annotations; those corrections do not constitute an independent audit of our constructed tasks. Public benchmark exposure during model training is unknown.

Evidence layer	Included here	What it establishes
Source utterances	Released dialogue prefixes with speaker and turn IDs	What the source user or assistant said
Source annotations	Corrected belief states, evaluator-only	The benchmark’s current constraints
Local database	Pinned hotel and restaurant records	Reproducible shortlist queries
Authored application	Two source epochs, paraphrased requests, up to six agent roles	A controlled multi-agent adaptation
Generated evidence	Qwen extraction, request parsing, full-history and memory answers	Fallible language interpretation
Simulated responses	Accurate answers only after explicit questions, within a response budget	Conditional task quality and additional question demand
Interface observations	Scripted browser actions and screenshots	Working software, not participant behavior

Roles prepare lodging and dining shortlists, a price-category brief, a location brief, an itinerary specification and accommodation requirements. These are distinct artifacts of **one source project**, not independent user goals. Queries execute against the local databases. Briefs reflect their consumed constraints. An empty correctly constrained shortlist is a completed local artifact, not a successful trip or booking. No purchase or booking is performed.

The frozen screen requires at least four supported final fields across lodging and dining. Only seven declared attributes are used. User-text lexical support is required, including explicit star phrases rather than a bare number. Unsupported, unstated and no-preference fields are excluded by a rule fixed before model outcomes. A lexical match is not proof of correct domain attribution or entailment.

The main development batch uses eight official validation dialogues. One extra validation dialogue appeared in the initial probe, so **nine distinct dialogues were used anywhere in development**. Evaluation uses 24 different official test dialogues, with the first 12 eligible IDs from each of the retained-change and stable strata. No generated outcome determined inclusion. The source-ID/text split check found no exact duplicates. A later source audit found 32 different hotel/restaurant goal signatures across the main development and evaluation selection. Including the extra probe dialogue gives 33 distinct goal signatures across all development/evaluation source IDs. Shared database records, task templates and unidentified source participants still limit independence. Related role variants stay within the same dialogue unit.

Declared comparison

The method, tasks, challenges, prompts, settings, source files and analysis were frozen in commit 70299a02 at **08:05:47 UTC**, before evaluation generation began at 08:06 UTC. The declaration and 21 checked hashes are under `artifacts/decision_dependencies/frozen/`.

The final matrix contains 24 dialogues, two independently seeded preparation replicates, role caps 2/4/6, response budgets 0/2/6/unlimited, and nine methods. That is **5,184 paired controller runs**, generated from 48 shared preparations and 576 final-dialogue calls. At the six-role cap there are 141 nonempty artifacts per replicate, or 282 per method. Three empty role instances are inactive and receive no success credit. No source case was dropped. Twelve separately authored scope challenges have two replicates and 96 calls. They are not additional source dialogues or human observations.

The structured methods share the same fallible extraction and request parsing. Full history answers from all released utterances. Semantic memory answers from all current extracted records. Per preparation, the common cache has two extractions, two request parsings, four full-history answers and four memory answers. The methods reuse those outputs across role and response-budget settings. This holds generated inputs fixed; it does not pretend that each controller row is a fresh agent trajectory. Baselines differ in the calls their mechanisms need, and the complete shared call cost is reported rather than charged repeatedly.

An inserted simulated answer is shared within its source epoch. A new source prefix invalidates temporary answers conservatively. We do not generate an alternative historical conversation after an inserted answer. This choice can create unnecessary repeat questions, and limits claims about continuous real use. The prototype’s explicit online overrides are separately tested user events. No operational deadlines or user response durations are invented for this task. Completion refers to release of checked local artifacts, not elapsed human time. Deadline misses and experienced workload therefore have no measured values here.

The co-primary descriptive outcomes are whole-project correctness and additional questions for selective checking minus semantic memory, at six roles with unlimited responses. All quantities are kept separate. Replicates are averaged within dialogue before 2,000 paired bootstrap resamples with seed 91831, stratified by retained change. All matched conditions use the same resampled dialogue indices. The sample was deliberately balanced for change, so these intervals describe a small selected sample, not the natural frequency of instruction changes.

Mechanism and information boundary

A candidate record contains a project, domain, attribute, value, user-source turn and quote, inferred or confirmed status, version and explicit exceptions. A separate model call interprets request scope from its wording. A source guard checks the schema, user role, turn and quotation. Explicit conditional phrases cause abstention. It is a conservative lexical rule, not a semantic scope proof. Automatic extraction represents scalar domain/attribute values; fine-grained exceptions must be recognized as uncertain or supplied explicitly by the user.

Each artifact registers the decision versions it consumed. A changed or revoked record invalidates dependent work. Selective checking rereads stale dependencies before release. The global baseline rereads all current inputs after every source epoch. Both should produce identical final values under the same records and deterministic artifact functions. A simpler read-only record baseline deliberately has no release check, exposing the established stale-cache failure.

The guarantee concerns stored versions and registered dependencies. It does not repair a misunderstood instruction, discover an omitted dependency or prevent an integrator from bypassing the release check. An actual quotation can support an outdated interpretation. A preference is never general action approval. Approval requests use a common request-specific guard in all challenge methods, so those challenge successes are credited to an automatic safeguard, not a unique advantage of the proposed method.

Only released source text, public project context and current requests enter model calls. Target answers and future turns are withheld. The workload builder uses annotations to choose already stated required fields, as an explicit task construction step. The request parser does not receive those correct scope labels. Evaluator values enter the controller only as an explicitly requested simulated user answer. A regression check changes hidden answer values while the response budget is zero and confirms unchanged controller events and outputs.

Results

Quality and additional decisions

The table reports totals across 48 source-replicate preparations. **The statistical unit remains 24 dialogues.** All rows use the six-role cap and unlimited additional answers. Every method has 282 nonempty local artifacts to complete.

Method	Correct projects / 48	Correct artifacts / 282	Additional questions	Incorrect answer uses
Independent requests	48	282	1,138	0
Full history	38	253	39	47
Semantic memory	44	267	58	31
Records at read	23	215	86	77
Global version check	44	268	148	22
Selective dependencies	44	268	148	22
No source guard	44	267	105	25
No domain scope	43	264	127	34
Confirm every new record	48	282	410	0

Source: `interpretation/primary_totals.csv`. Incorrect answer uses include initial, final and retained uses, so they are not identical to incorrectly released artifacts. No unresolved artifacts remain in this unlimited-response reference. Finite-budget outcomes retain unresolved work below.

Selective checking and semantic memory have the same whole-project correctness for **all 24 paired dialogue means**. Their empirical difference and bootstrap interval are $[0, 0, 0]$. A degenerate interval reflects identical observed paired values; it is not proof that the methods are equivalent on future cases. The selective method asks 3.083 questions per project versus 1.208 for semantic memory. The paired difference is $+1.875 [1.041, 2.854]$, with fewer questions on 2 dialogues, a tie on 4 and more questions on 18.

Artifact accuracy differs by only $+0.347$ percentage points on the dialogue-mean scale, with a 95% interval $[0, 1.042]$. One dialogue favors selective checking and 23 tie. This small artifact difference disappears in the normalization sensitivity below. The outcome does not establish an automatic reuse advantage.

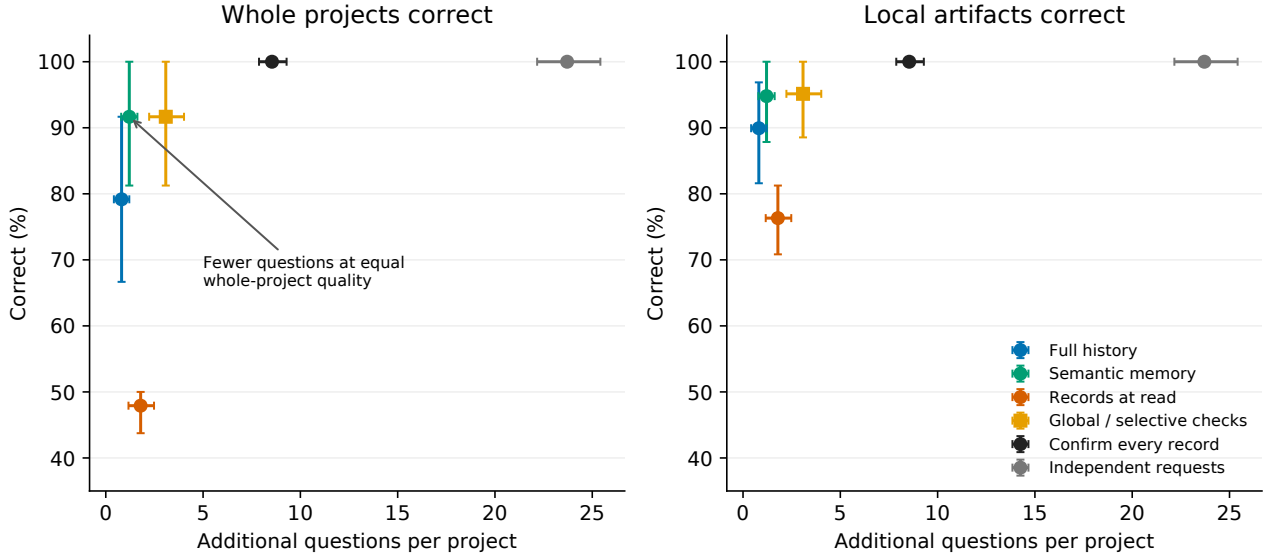


Figure 1: Generated interpretations and simulated additional responses on 24 source dialogues. Two replicates are averaged within dialogue. Bars are dialogue-level bootstrap intervals. The axes show distinct outcomes, not a combined utility or observed mental workload. Independent asking is dominated in the plotted means by confirmed sharing; other methods trade quality against questions. The vertical axis begins at 35 percent for readability.

Demand and limited responses

With only two additional answers available per project, semantic memory completes 39/48 whole projects and 250/282 artifacts correctly. Selective checking completes 25/48 projects and 193/282 artifacts. It leaves 81 artifacts unresolved, versus 17 under semantic memory, and releases 8 incorrect artifacts versus 15. Its paired whole-project difference is -29.17 percentage points, with a 95% interval [-47.92, -10.42]. The controller’s caution lowers some wrong completions but consumes scarce responses and leaves substantially more work unfinished.

With six responses available, selective checking still leaves nine artifacts unresolved and completes 40/48 projects, while semantic memory already reaches its unlimited-response outcome of 44/48. Confirming every new record attains perfect quality only in the unlimited-response reference. At a six-answer budget it leaves 154/282 artifacts unresolved. Suppressing wrong releases through excessive confirmation is not free progress.

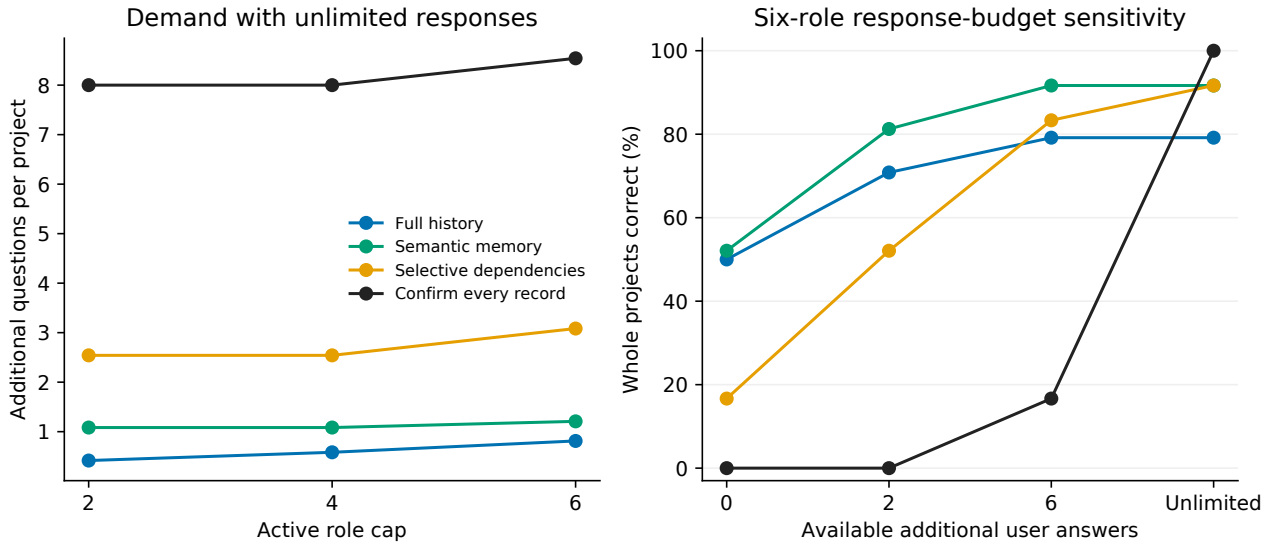


Figure 2: Constructed role counts and response budgets applied to identical saved outputs. The budget counts additional simulated answers, not source utterances or seconds. Whole-project correctness includes unresolved work as failure. Global checks coincide with selective checks and are omitted as a duplicate curve.

The distinct-question accounting also rejects a superficial sharing claim. Selective checking asks for 98 distinct underlying key/value decisions and repeats 50 of them across the 48 preparations. Semantic memory asks for

49 distinct decisions and repeats nine. Both substantially reduce the independent-asking reference, but the semantic baseline is more frugal. One answer is counted once only when its explicit scope actually permits reuse. A display containing several separate questions does not receive a one-decision discount.

What version checks contribute

Records without a release check recover none of the 58 changed-field uses across the source-replicate preparations. Selective, global and semantic-memory methods recover all 58 in the unlimited reference. Full history recovers 54. These counts refer to retained benchmark changes between the selected epochs, not every earlier change anywhere in the conversation.

Global and selective checks have identical artifacts, questions and answered questions in **every matched demand/budget/replicate condition**, verified directly from traces. Selective checking reduces repeat value reads from 462 to 261, a 43.5% reduction. These are cheap host operations. The full controller simulations, including local artifact scoring, have median 1.87 ms and p95 2.99 ms per row on this host. There is no measured human-reading saving to offset the additional questions versus semantic memory.

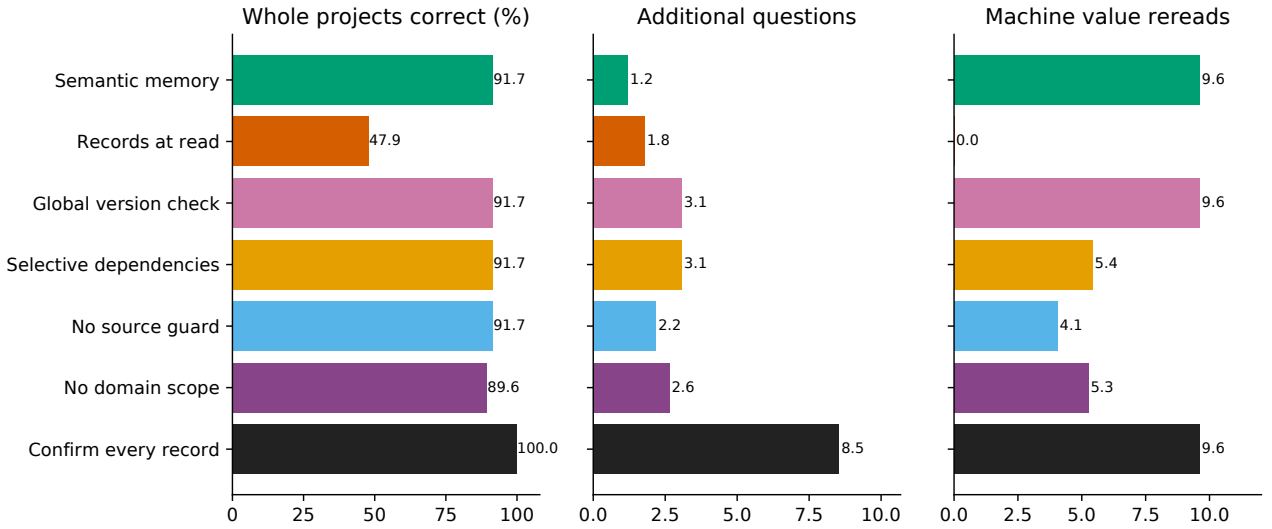


Figure 3: Frozen ablations with six roles and unlimited accurate responses. Questions and machine rereads are per-project means. The simple global check matches selective quality and question demand. Removing domain scope worsens correctness. Removing source checks asks fewer questions with almost unchanged quality. These are generated/model and simulated/controller outcomes.

The declared inspection sensitivity assigns 0, 0.25 or 1 additional assumed inspection equivalent per distinct candidate source, without claiming people actually inspect each card. At zero cost, selective checking needs 3.083 units versus 1.208 for semantic memory. At 0.25 it needs 4.729 versus 3.328. At one it needs 9.667 versus 9.688, essentially tied. A presumed large inspection saving can erase the question disadvantage, but this is an assumption requiring actual interaction evidence. It is not an observed advantage.

Language interpretation and propagated mistakes

The request parser correctly maps 624/672 public field queries and abstains on 48. Every abstention concerns the first accommodation-type paraphrase. Structured methods therefore face a systematic applicability gap even though the question is simple. Raw records supply 328/384 evaluated state fields, with nine incorrect values. Source checks reduce supply to 269/384, with six incorrect values. Correct coverage falls from 319/384 to 263/384. The resulting precision change is small relative to the coverage loss. A source quote alone does not solve stale meaning.

The two generation replicates produce different guarded states on 10/24 source dialogues, differing on 23 of 145 compared field positions. This is sampling variation, not a repeated-identical-seed reproducibility claim. Policy comparisons use the same saved replicate. Deterministic replay does not imply deterministic fresh generation.

At the six-role unlimited setting, all structured/semantic-memory project failures are concentrated in three source dialogues. Two contain the phrase **center of town**, which the frozen adapter fails to map to its database value **centre**. The remaining dialogue, MUL1376, changes from an expensive hotel to a cheap guesthouse. The

extractor retains the original expensive price. Both replicates propagate this stale preference into four derived artifacts. Full-history and memory systems also have their own interpretation errors; no method’s outputs were substituted or repaired in the primary tables.

A **post hoc hypothetical adapter repair** maps only *.area: center of town to centre, recomputes local artifacts, and preserves every controller decision and question. Semantic memory and both version checks then each complete 46/48 projects and 274/282 artifacts correctly. The remaining eight incorrect artifacts are the genuinely stale-preference failures. This sensitivity eliminates the one-artifact selective advantage, while retaining its 90 additional questions. It does not replace the frozen evaluation or constitute a fresh repaired run.

Exceptions, revocation and authority

The twelve authored cases probe ordinary sharing, domain differences, revisions, conflict, approvals, exceptions, another project’s instruction, agent inference, revocation and an explicit cross-domain reference. They are a separate controlled layer and receive no population interval.

Method	Correct automatic answers	Appropriate abstentions	Wrong transfers	Unnecessary questions
Full history	14	10	0	0
Semantic memory	12	10	0	2
Global check	10	8	4	2
Selective check	10	8	4	2

Each row represents 12 authored cases times two generations. A revocation and a paraphrased workshop-evening exception fail in both replicates under the structured methods. The earlier cheap value has a real source quotation, but is no longer the correct answer to the current question. Full history handles all these cases. This is direct evidence against equating explicit record fields with successful language-level scope recognition.

Authored scope challenges: two generations per case				
ordinary	2/2 valid	2/2 valid	2/2 valid	2/2 valid
different domains	2/2 valid	2/2 valid	2/2 valid	2/2 valid
revision	2/2 valid	2/2 valid	2/2 valid	2/2 valid
undecided conflict	2/2 valid	2/2 valid	2/2 valid	2/2 valid
preference not approval	2/2 valid	2/2 valid	2/2 valid	2/2 valid
specific approval	2/2 valid	2/2 valid	2/2 valid	2/2 valid
explicit exception	2/2 valid	2/2 valid	2/2 asks	2/2 asks
paraphrased exception	2/2 valid	2/2 asks	2/2 wrong	2/2 wrong
similar project	2/2 valid	2/2 valid	2/2 valid	2/2 valid
inference not instruction	2/2 valid	2/2 valid	2/2 valid	2/2 valid
revocation	2/2 valid	2/2 valid	2/2 wrong	2/2 wrong
explicit reference	2/2 valid	2/2 valid	2/2 valid	2/2 valid
	Full history	Semantic memory	Global check	Selective check

Figure 4: Generated interpretations of twelve authored cases, with two replicates per case. Green includes correct automatic answers and appropriate abstentions. Yellow is an extra question when an answer exists. Red is an incorrect transfer. The common approval guard is enabled in every method. These are not participant responses or independent MultiWOZ dialogues.

Readable benefit, tie and failure

The first qualifying benefit over read-only records is MUL0071, replicate 0. Selective checking releases all six artifacts correctly after the source update, with one extra question. The read-only baseline asks no questions but releases four incorrect artifacts. The same benefit is supplied by the global check and does not distinguish a new algorithm.

The first quality tie with greater question demand is MUL0003, replicate 0. Semantic memory asks one question, while selective checking asks seven. Both release all six artifacts correctly. This is a concrete unfavorable demand example. There is no qualifying unlimited-response artifact-quality loss against semantic memory in the frozen selection rule, and that absence is reported rather than replaced by a more dramatic case.

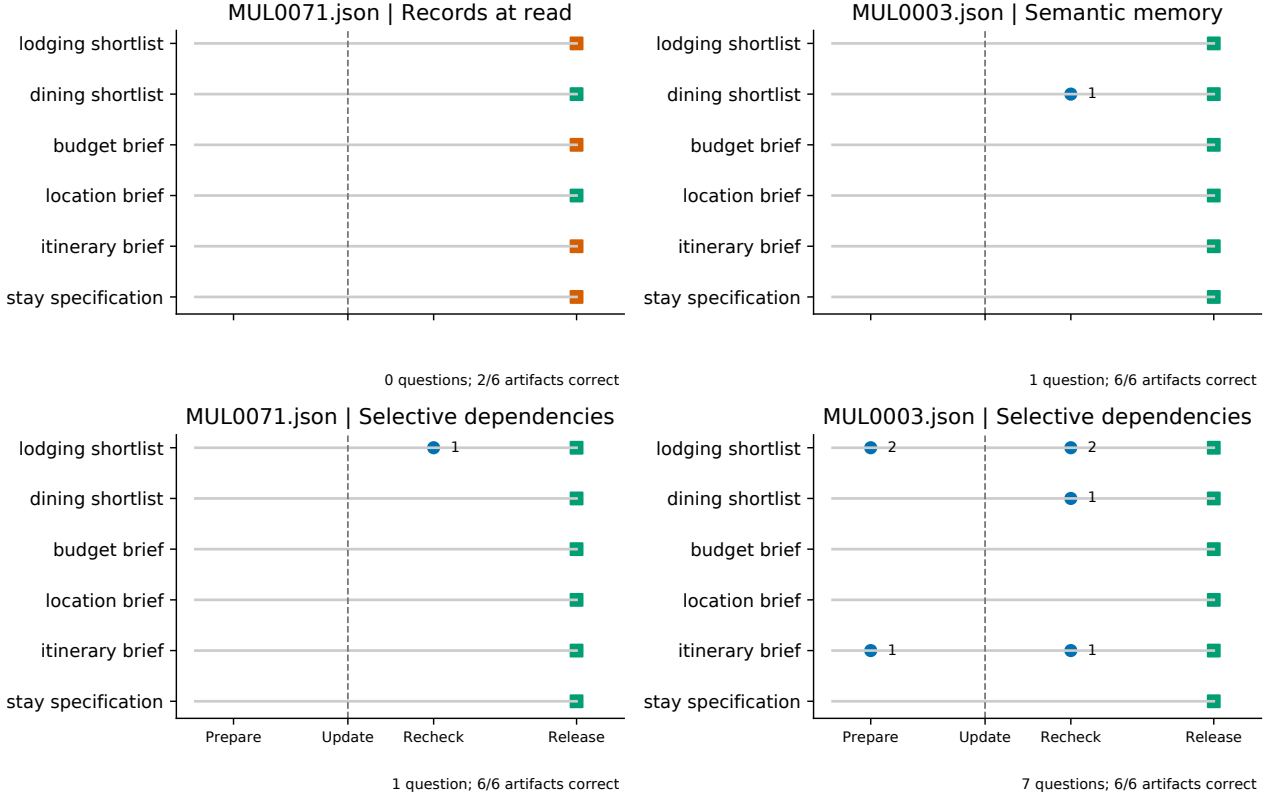


Figure 5: Logical phases and per-role question counts on matched saved inputs. Source utterances are task-elicited, interpretations generated, and role execution simulated. Green/red squares indicate correct/incorrect local releases. Horizontal spacing is not measured user response time. Examples use first-qualifying source IDs, not the largest effect.

The explicit failure walkthrough is the authored revocation case. The source user withdraws a cheap-hotel preference and supplies no replacement. The extractor returns the old cheap value. The version controller consistently propagates that incorrect stored interpretation. A user pressing the interface’s **Revoke** button does invalidate registered work correctly. Those are different tests: explicit structured revocation works, automatic understanding of a revocation failed.

Working prototype and verification

The interface shows pending questions, candidate earlier answers, provenance, inferred versus confirmed scope, affected work, deferrals and explicit overrides. It distinguishes request-only answers from named shared scope. A conference-night exception cannot silently inherit or widen the ordinary answer. Unknown scope stays unresolved until the user supplies it. Changed instructions mark stale drafts, and the release endpoint enforces revalidation. Already released events remain in the audit trail.

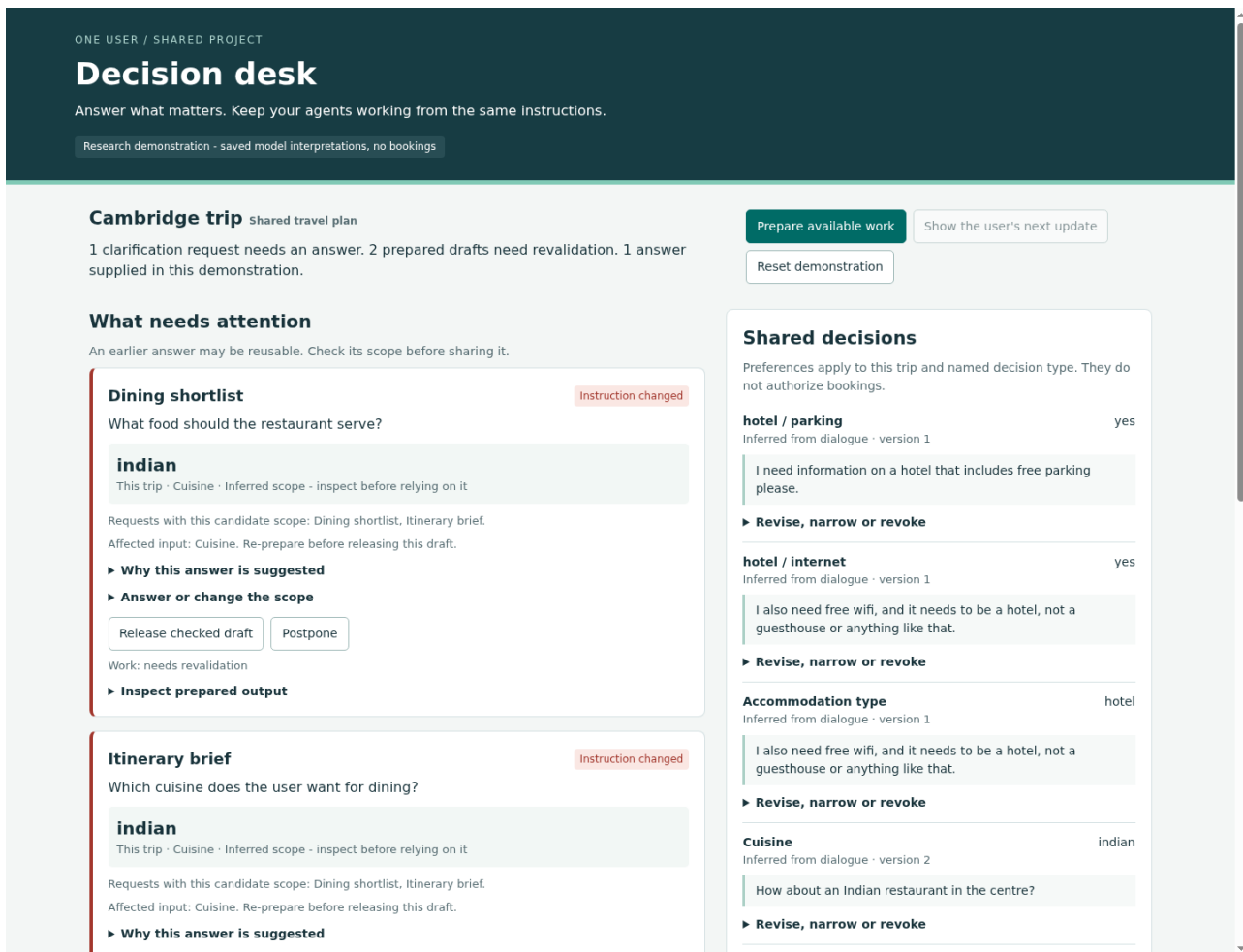


Figure 6: A scripted browser demonstration using saved development outputs. It records no participant behavior. The interface deliberately shows unresolved extraction failures and affected drafts.

Run from the repository root:

```
python3 research/decision_dependencies/prototype/server.py --port 9027
```

Open <http://127.0.0.1:9027>. The [walkthrough and Python adapter](#) show how several roles submit requests, consume a scoped answer and check work release. The interface uses saved development outputs, not new inference. External agents may supply candidate scope, which remains a hypothesis. The API does not silently run a model for arbitrary new text or authorize external actions.

Interaction logs record displayed cards, source inspection, answers, scope choices, changes, rejected operations, deferrals, overrides and time since a card was shown. Those scripted timings are not reading time or mental workload. The [human-study protocol](#) compares the same strong shared-memory and consistency backend with or without explicit scope/change receipts. Its eventual participant sample size must come from actual pilot variability and a declared precision target, not from the 24 source dialogues here.

Twenty-four focused tests pass. They cover label isolation, source boundaries, exceptions, approval scope, explicit revocation, stale release handling, online override atomicity, response budgets, attempt/cutoff limits, global/selective equivalence and raw accounting. The browser walkthrough passes actual form submission, shared answers, source changes, stale-release rejection, independent exceptions, deferral, mobile overflow and no JavaScript exceptions. Historical attention-session tests passed (17), and all 18,759 files tracked at the starting commit remain unchanged.

Saved-output reproduction checks all 5,184 canonical traces, episode CSV and five frozen analysis tables. It reconstructs 56 main development/evaluation preparations and 24 challenge preparations from raw responses, checks 768 complete prompt/seed/setting payloads, and replays all 120 challenge comparison rows. The 16 earlier probe calls remain fully accounted raw evidence. A separate download audit reconstructs the pinned source selection and verifies exact selected bytes.

One command reproduces results and figures without a GPU or network:

```
python3 -m research.decision_dependencies.reproduce --figures
```

It writes to `/tmp/decision-dependencies-replay`, preserving original evidence. The five report figures have vector PDF/SVG and 300 dpi PNG exports, source hashes and captions under `publication/`. Frozen diagnostic figures also remain in `evaluation/figures/`. The post hoc area-alias and inspection-cost analyses are explicitly separate in `interpretation/`.

Resources and limits

The separately authorized session began **2026-09-11 07:16:21 UTC**, with a hard stop at 10:16:21 UTC and an inference cutoff at 09:46:21 UTC. Actual generation finished at **08:33:00 UTC**. No new resource was provisioned. The existing RTX 6000 Ada and vLLM server were reused with pinned Qwen2.5-7B-Instruct revision `a09a35458c702b33eeacc393d103063234e8bc28`, BF16, one GPU, zero CPU offloading and serial requests. Temperature is 0.3, top-p 1.0, context limit 2,048, and output limits 768 for extraction and 384 for other calls. JSON object formatting is a separately recorded application setting. The earlier unconstrained failures remain.

Phase	Calls / attempts	Prompt tokens	Completion tokens
Development and format probes	112 / 112	59,419	14,544
Final dialogue preparations	576 / 576	286,180	60,449
Authored scope challenges	96 / 96	17,460	1,912
Total	784 / 784	363,059	76,905

There were zero transport failures, zero retries and zero context-limit rejections. Three responses were unparseable. One final extraction hit its 768-token limit; two original development probe outputs were malformed despite normal stop reasons. Every failed output remains in the raw data and dependent accounting. Final dialogue calls averaged 2.30 seconds, with interpolated p95 4.10 seconds. The sum of generation-attempt durations is 1,699.43 seconds, or 28m 19s. This is measured request duration, not a billed allocation measurement or user inspection time.

Server counter differences match all **784 requests, 363,059 prompt tokens and 76,905 completion tokens exactly**. The existing model server remains healthy and idle; the temporary SSH tunnel was stopped after collection. Final local process cleanup and session elapsed time are recorded in `resource_ledger.json`. Provider billing rate and remaining allocation balance were unavailable.

Cumulative available accounting is 56,370 scheduled calls, 56,372 attempts, 31,159,693 prompt tokens and 761,833 completion tokens. The original start and 36-hour deadline remain 9 September 15:38:57 UTC and 11 September 03:38:57 UTC. This new session occurs **outside that original window under new authorization**. The final ledger reports wall time since the original start, including inter-session gaps, and the overrun explicitly. It does not reset historical clocks or describe new work as part of the original 36 hours.

Recommendation

Keep the working decision desk and simplify the automatic backend around a strong shared-memory or full-history interpreter with a global change barrier. Retain explicit user scope, request-only answers, revocation and visible affected work. Selective dependencies can remain an optional engineering optimization if larger artifacts make rereads expensive. This pilot does not justify them as an attention or quality improvement.

The specific research lesson is that **consistency of a stored instruction is different from correctness of its interpretation**. Conservative extraction can increase supervisory demand enough to reduce completed work when answers are scarce. Familiar memory and a simple consistency check are essential comparators. The clearest next test is whether explicitly capturing a user's intended scope and showing change-impact receipts improves real interaction over the same competent backend. The current package makes that test possible without claiming it has already measured comprehension, control or cognitive load.

Treat this direction as follow-up work. Keep the completed ICAART submission intact. Its unfavorable retail results, the HVAC primary mechanical tie, harmful model review, ideal EDF/search tie and lower greedy review count remain unchanged. They motivate attention-value questions but do not validate the new interaction.